Hence, $Q = +6\ J$

18 Ans: C

$W = (2.6 \times 10^{-15})\ 9.81$
$W = 2.55 \times 10^{-14}$ N

Electric Field strength between plates, $E = V / d = 1200 / 0.02 = 60\ 000$ V/m

Upward Electric Force, $F = qE = (4.8 \times 10^{-19})\ 60\ 000 = 2.88 \times 10^{-14}$ N

$F_{net} = F - W$
$ma = (2.88 - 2.55) \times 10^{-14}$
$a = (0.33 \times 10^{-14}) / (2.6 \times 10^{-15})$
$a = 1.27 \approx 1.3\ ms^{-2}$

10 Ans: B